

UNIT - 03 Interpolation (प्रक्षेपण)

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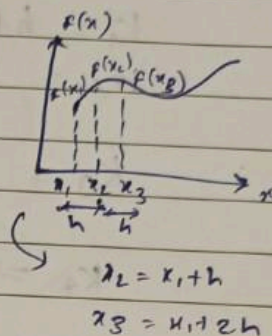
* Finite Difference Techniques :-

① Shift operator / Increment operator :

$$E = f(x+h)$$

$$E y_x = y_{(x+h)}$$

$$E^{-1} y_{(x+h)} = y_x$$



② Forward difference operator (Δ) :

consider a function $y = f(x)$, suppose we have given values of the function at the point $x_0, x_0+h, x_0+2h, \dots, x_n = x_0+nh$.
Let $f(x_0) = y_0, f(x_1) = y_1, f(x_2) = y_2, \dots, f(x_n) = y_n$.

we define, $\Delta[f(x)] = f(x+h) - f(x)$

$$\begin{aligned} \text{Thus, } \Delta y_0 &= f(x_0+h) - f(x_0) \\ &= y_1 - y_0 \end{aligned}$$

Here $x_1, x_2, x_3, \dots, x_n$ are called arguments corresponding $f(x_1), f(x_2), f(x_3), \dots, f(x_n)$ are called entries & h is called interval of differences.

$$\Delta y_1 = y_2 - y_1$$

$$\Delta y_2 = y_3 - y_2$$

$$\Delta y_{n+1} = y_{n+1} - y_n$$

Δ is called forward difference operator and $\Delta y_1, \Delta y_2, \dots, \Delta y_n$ are called first forward differences of function $y = f(x)$.

Second order differences of $y = f(x)$ are:

$$\Delta^2 y_0 = \Delta y_1 - \Delta y_0$$

$$\Delta^2 y_1 = \Delta y_2 - \Delta y_1$$

$$\Delta^2 y_n = \Delta y_{n+1} - \Delta y_n$$

Similarly, $\Delta^2 y_0 = \Delta y_1 - \Delta y_0$

$$= y_2 - y_1 - (y_1 - y_0)$$

$$= y_2 - y_1 - y_1 + y_0$$

$$\boxed{\Delta^2 y_0 = y_2 - 2y_1 + y_0}$$

Also, $\Delta^3 y_0 = \Delta^2 y_1 - \Delta^2 y_0$

$$= (y_3 - 2y_2 + y_1) - (y_2 - 2y_1 + y_0)$$

$$\Delta^3 y_0 = y_3 - 3y_2 + 3y_1 - y_0 \rightarrow (1-x)^n$$

* coefficient occurring in RHS are simply binomial coefficient in $(1-x)^n$. In general,

$$\Delta^n y_0 = y_n - {}^nC_1 y_{n-1} + {}^nC_2 y_{n-2} - {}^nC_3 y_{n-3} + \dots + (-1)^n y_0$$

Systematic representation Forward difference table.

x	$y = f(x)$	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$	$\Delta^5 y$
x_0	y_0					
x_0+h	y_1	$y_1 - y_0 = \Delta y_0$				
x_0+2h	y_2	$y_2 - y_1 = \Delta y_1$	$\Delta y_1 - \Delta y_0 = \Delta^2 y_0$			
x_0+3h	y_3	$y_3 - y_2 = \Delta y_2$	$\Delta y_2 - \Delta y_1 = \Delta^2 y_1$	$\Delta^2 y_1 - \Delta^2 y_0 = \Delta^3 y_0$		
x_0+4h	y_4	$y_4 - y_3 = \Delta y_3$	$\Delta y_3 - \Delta y_2 = \Delta^2 y_2$	$\Delta^2 y_2 - \Delta^2 y_1 = \Delta^3 y_1$	$\Delta^3 y_1 - \Delta^3 y_0 = \Delta^4 y_0$	
x_0+5h	y_5	$y_5 - y_4 = \Delta y_4$	$\Delta y_4 - \Delta y_3 = \Delta^2 y_3$	$\Delta^2 y_3 - \Delta^2 y_2 = \Delta^3 y_2$	$\Delta^3 y_2 - \Delta^3 y_1 = \Delta^4 y_1$	$\Delta^4 y_1 - \Delta^4 y_0 = \Delta^5 y_0$

Forward difference formula:

$$p = \frac{x - x_0}{h}$$

$$y(x) = y_0 + p \Delta y_0 + \frac{p(p-1)}{2!} \Delta^2 y_0 + \frac{p(p-1)(p-2)}{3!} \Delta^3 y_0 + \dots$$

*

Properties of Forward shift operator:

- (1) Δ is linear, $\Delta[af(x) + bg(x)] = a\Delta f(x) + b\Delta g(x)$
- (2) $\Delta C = 0$, C is being constant.
- (3) $\Delta^n \Delta^m f(x) = \Delta^{n+m} f(x)$
- (4) $\Delta[f(x) \cdot g(x)] = f(x+h)\Delta g(x) + g(x)\Delta f(x)$
- (5) $\Delta \left[\frac{f(x)}{g(x)} \right] = \frac{g(x)\Delta f(x) - f(x)\Delta g(x)}{g(x+h)g(x)}$

(3) Backward difference operator (∇):

$$\nabla f(x) = \cancel{f(x-h)} - f(x) \quad f(x) - f(x-h)$$

$$\nabla y_0 = y_0 - y_{-1}$$

$$\nabla y_1 = y_1 - y_0$$

$$\nabla y_n = y_n - y_{n-1}$$

second order: $\nabla^2 y_2 = \nabla y_2 - \nabla y_1$

$$\nabla^3 y_2 = \nabla^2 y_2 - \nabla^2 y_1$$

x	$y = f(x)$	∇y	$\nabla^2 y$	$\nabla^3 y$	$\nabla^4 y$	$\nabla^5 y$
x_0	y_0	$\Delta y_1 = y_1 - y_0 = y_{01}$	$y_2 - y_0 = y_{201}$	$y_3 - y_0 = y_{301}$	$\Delta^4 y_4$	$\Delta^5 y_5$
x_0+h	y_1					
x_0+2h	y_2	$\Delta y_2 = y_2 - y_1 = y_{21}$	$y_3 - y_2 = y_{32}$	$\Delta^3 y_4$	$\Delta^4 y_5$	$\Delta^5 y_6$
x_0+3h	y_3	$\Delta y_3 = y_3 - y_2 = y_{31}$				
x_0+4h	y_4	$\Delta y_4 = y_4 - y_3 = y_{41}$	$y_5 - y_4 = y_{54}$	$\Delta^3 y_5$	$\Delta^4 y_6$	$\Delta^5 y_7$
x_0+5h	y_5	$\Delta y_5 = y_5 - y_4 = y_{51}$				

Backward difference formula:

$$p = \frac{x - x_0}{h}$$

$$f(x) = y_n + p \nabla y_n + \frac{p(p+1)}{2!} \nabla^2 y_n + \frac{p(p+1)(p+2)}{3!} \nabla^3 y_n + \dots$$

20) Find the relation of the given data:
using forward diff. formula

x	1891	1901	1911	1921	1931	1895
f(x)	46	66	81	92	101	?

$x = 1895$, let $x_0 = 1891$, $h = 10 \Rightarrow p = \frac{1895 - 1891}{10} = 0.4$

x	f(x)	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$
1891	46	20	-5	+2	-3
1901	66	15	-3	-1	
1911	81	12	-4		
1921	92	8			
1931	101				

$$f(1895) = y_0 + p \Delta y_0 + \frac{p(p-1)}{2!} \Delta^2 y_0 + \frac{p(p-1)(p-2)}{3!} \Delta^3 y_0$$

$$+ \frac{p(p-1)(p-2)(p-3)}{4!} \Delta^4 y_0 + \dots$$

$$= 46 + 0.4(20) + \frac{(0.4 \times -0.6)}{2} (-5) + \frac{(0.4 \times -0.6 \times -1.6)}{6} (-3)$$

$$= 46 + 8 - 0.6 + 0.128 + 0.1299$$

$$f(1895) = 54.9588$$

Q.9) Same table but $x = 1925$?

Table with 8 columns but we will consider values from row 1.

x	$f(x)$	∇y	$\nabla^2 y$	$\nabla^3 y$	$\nabla^4 y$
x_0 1891	46				
x_1 1901	66	$\nabla y_0 = 20$	$\nabla^2 y_0 = -5$	$\nabla^3 y_0 = 2$	$\nabla^4 y_0 = -3$
x_2 1911	81	$\nabla y_1 = 15$	$\nabla^2 y_1 = -3$	$\nabla^3 y_1 = -1$	
x_3 1921	93	$\nabla y_2 = 12$	$\nabla^2 y_2 = -4$		
x_4 1931	101	$\nabla y_3 = 8$			

$$p = \frac{x - x_0}{h} = \frac{1925 - 1931}{10} = \frac{-6}{10} = -0.6$$

$$f(1925) = y_4 + p \nabla y_3 + \frac{p(p+1)}{2!} \nabla^2 y_3 + \frac{p(p+1)(p+2)}{3!} \nabla^3 y_3 + \frac{p(p+1)(p+2)(p+3)}{4!} \nabla^4 y_3$$

$$= 101 + (-0.6) \cdot 8 + \frac{(-0.6)(-0.6+1)(-0.6+2)}{2!} (-4) + \frac{(-0.6)(-0.6+1)(-0.6+2)(-0.6+3)}{3!} (-1)$$

$$= 101 - 4.8 + \frac{(-0.6)(0.4)(1.4)}{2} (-4) + \frac{(-0.6)(0.4)(1.4)(2.4)}{6} (-1)$$

$$= 101 - 4.8 + 0.48 + 0.064 = 101.156$$

$$\Rightarrow 101.156$$

$$f(1925) = 96.8388$$

* Gauss forward:

$$f(a+hu) = y_0 + u \Delta y_0 + \frac{u(u-1)}{2!} \Delta^2 y_{-1} + \frac{(u+1)u(u-1)}{3!} \Delta^3 y_{-1} + \frac{(u+1)u(u-1)(u-2)}{4!} \Delta^4 y_{-2}$$

* Gauss backward:

$$f(a+hu) = y_0 + u \Delta y_{-1} + \frac{u(u+1)}{2!} \Delta^2 y_{-1} + \frac{(u-1)u(u+1)}{3!} \Delta^3 y_{-2} + \frac{(u-1)u(u+1)(u+2)}{4!} \Delta^4 y_{-2}$$

Q.1)

Find $f(9)$?

x	f(x)	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$
0	14	y_{-2}			
4	24	y_{-1}	$10(\Delta y_{-2})$		
8	32	y_0	$-2(\Delta^2 y_{-2})$	$8(\Delta y_{-1})$	$-3(\Delta^3 y_{-2})$
12	35	y_1	$-5(\Delta^2 y_{-1})$	$3(\Delta y_0)$	$10(\Delta^4 y_{-2})$
16	40	y_2	$2(\Delta^2 y_0)$	$7(\Delta^3 y_{-1})$	
			$5(\Delta y_1)$		

By Gauss forward:

$$f(a+hu) = y_0 + u \Delta y_0 + \frac{u(u-1)}{2!} \Delta^2 y_{-1} + \frac{(u+1)u(u-1)}{3!} \Delta^3 y_{-1} + \frac{(u+1)u(u-1)(u-2)}{4!} \Delta^4 y_{-2}$$

Here $a=8, h=4, \Rightarrow 8+4u=9$
 $\Rightarrow u=0.25$

$$F(9) = 32 + (0.25)3 + \frac{(0.25)(-0.75)(-5)}{2} + \frac{(1.25)(0.25)(-0.75)}{6} \times 7$$

$$+ \frac{(1.25)(0.25)(-0.75)(-1.75)}{24} \times 10$$

$$F(9) = 33.1162$$

By Gauss backward:

$$F(a+hu) = y_0 + u\Delta y_1 + \frac{u(u+1)}{2!}\Delta^2 y_{-1} + \frac{(u+1)u(u+1)}{3!}\Delta^3 y_{-2}$$

$$+ \frac{(u+1)u(u+1)(u+2)}{4!}\Delta^4 y_{-2}$$

again $u=0.25$

$$F(9) = 32 + (0.25)8 + \frac{(0.25)(-0.75)(-5)}{2}$$

$$+ \frac{(-0.75)(0.25)(1.25)(-3)}{6} + \frac{(-0.75)(0.25)(1.25)(2.25)}{24} \times 10$$

$$F(9) = 33.1162$$

* Stirling formula:

$$F(a+hu) = y_0 + u(\Delta y_0 + \Delta y_{-1}) + \frac{u^2}{2!}\Delta^2 y_{-1} + \frac{u(u^2-1)}{3!}\left(\frac{\Delta^3 y_{-1} + \Delta^3 y_{-2}}{2}\right)$$

$$+ \frac{u^4(u^2-1)}{4!}\Delta^4 y_{-2}$$

$$f(x, x_1) = [x, x_1] = \frac{y_0 - y_1}{x_0 - x_1} = \frac{f(x_0) - f(x_1)}{x_0 - x_1}$$

where,

$$[x_1, x_2, x_3] = \frac{[x_2, x_3] - [x_1, x_2]}{x_3 - x_1}$$

* Newton's dividend difference :

$$f(x) = f(x_0) + (x-x_0)f(x_1, x_0) + (x-x_0)(x-x_1)f(x_2, x_1, x_0) + (x-x_0)(x-x_1)(x-x_2)f(x_3, x_2, x_1, x_0) + \dots$$

Q.0) Find $f(2.5)$

x	$f(x)$	First diff	second diff	third	fourth
0	1				
		$\frac{14-1}{1-0} = 13$			
1	14		$\frac{1-13}{1-0} = -6$		
		$\frac{15-14}{2-1} = 1$		$\frac{-2+6}{2-1} = 1$	
2	15		$\frac{-5-1}{4-1} = -2$		0
		$\frac{5-15}{4-2} = -5$		$\frac{2+2}{5-1} = 1$	
4	5		$\frac{1+5}{5-2} = 2$		0
		$\frac{6-5}{5-4} = 1$		$\frac{6-2}{6-2} = 1$	
5	6		$\frac{3-1}{6-4} = 1$		
		$\frac{19-6}{6-5} = 13$			
6	19				

$$f(2.5) = 1 + (2.5-0)(13) + (2.5-0)(2.5-1)(-6) + (2.5)(1.5)(0.5)(1) = 12.875$$

Qo) Find $f(2)$?

x	$f(x)$	First diff.	Second diff.	third.
0	1	$\frac{3-1}{1-0} = 2$		
1	3	$\frac{55-3}{3-1} = 26$	$\frac{26-2}{3-0} = 8$	$\frac{14-8}{6} = 1$
3	55	$\frac{343-55}{6-3} = 96$	$\frac{96-8}{6-1} = 14$	
6	343			

$$f(x) = f(x_0) + (x-x_0)f(x_1, x_0) + (x-x_0)(x-x_0)f(x_2, x_1, x_0) + (x-x_0)(x-x_1)(x-x_2)f(x_3, x_2, x_1, x_0)$$

$$f(2) = 1 + (2-0)(2) + (2)(1)(8) + (2)(1)(-1)(1)$$

$$= 1 + 4 + 16 - 2$$

$$f(2) = 19$$

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* Interpolation with unequal Intervals:

→ Lagrange's Interpolation formula :-

If $f(x) = y$ the function such that
 $f(x_1) = y_1$, $f(x_2) = y_2$, $f(x_3) = y_3$ --- $f(x_n) = y_n$
 where x_1, x_2, x_3 --- x_n are the arguments
 having equispaced (unequal step size).

Lagrange's Formula $Y = \sum_{i=0}^n L_i y_i = L_0 y_0 + L_1 y_1 +$

--- $+ L_n y_n$.

$$L_0 = \frac{(x-x_1)(x-x_2) \dots (x-x_n)}{(x_0-x_1)(x_0-x_2) \dots (x_0-x_n)}$$

$$L_1 = \frac{(x-x_0)(x-x_2)(x-x_3) \dots (x-x_n)}{(x_1-x_0)(x_1-x_2)(x_1-x_3) \dots (x_1-x_n)}$$

$$L_2 = \frac{(x-x_0)(x-x_1)(x-x_3) \dots (x-x_n)}{(x_2-x_0)(x_2-x_1)(x_2-x_3) \dots (x_2-x_n)}$$

⋮

$$L_n = \frac{(x-x_0)(x-x_1)(x-x_2) \dots (x-x_{n-1})}{(x_n-x_0)(x_n-x_1)(x_n-x_2) \dots (x_n-x_{n-1})}$$

Q0) The given function, find $f(10) = ?$

	x_0	x_1	x_2	x_3
x	5	6	9	11
y	12	13	14	16

$$Y = \sum_{x=0}^3 L_i y_i = L_0 y_0 + L_1 y_1 + L_2 y_2 + L_3 y_3$$

$$= L_0(12) + L_1(13) + L_2(14) + L_3(16)$$

$$L_0 = \frac{(x-x_1)(x-x_2)(x-x_3)}{(x_0-x_1)(x_0-x_2)(x_0-x_3)} = \frac{(10-6)(10-9)(10-11)}{(5-6)(5-9)(5-11)}$$

$$= \frac{-4}{-4 \times 6} = \frac{1}{6}$$

$$L_1 = \frac{(x-x_0)(x-x_2)(x-x_3)}{(x_1-x_0)(x_1-x_2)(x_1-x_3)} = \frac{(10-5)(10-9)(10-11)}{(6-5)(6-9)(6-11)}$$

$$L_1 = \frac{5 \times 1 \times (-1)}{1 \times (-3) \times (-5)} = -\frac{1}{3}$$

$$L_2 = \frac{(x-x_0)(x-x_1)(x-x_3)}{(x_2-x_0)(x_2-x_1)(x_2-x_3)} = \frac{(10-5)(10-6)(10-11)}{(9-5)(9-6)(9-11)}$$

$$= \frac{5 \times 4 \times (-1)}{4 \times 3 \times (-2)} = \frac{5}{6}$$

$$L_3 = \frac{(x-x_0)(x-x_1)(x-x_2)}{(x_3-x_0)(x_3-x_1)(x_3-x_2)} = \frac{(10-5)(10-6)(10-9)}{(11-5)(11-6)(11-9)}$$

$$Y = 2 + \frac{13}{7} + \frac{15}{6}(14) + \frac{1}{3}(16)$$

$$Y = 19.6667$$

Q2) Compare the value of function at $F(1.5)$ for the given set of discrete data using NDD & LIF.

	x_0	x_1	x_2	x_3
x	-1	1	2	3
$F(x)$	-21	18	12	3

By Lagrange's :

$$Y = \sum_{i=0}^3 L_i y_i = L_0 y_0 + L_1 y_1 + L_2 y_2 + L_3 y_3$$

$$= L_0(-21) + L_1(18) + L_2(12) + L_3(3)$$

$$L_0 = \frac{(1.5-1)(1.5-2)(1.5-3)}{(-1-1)(-1-2)(-1-3)} = \frac{0.5 \times (-0.5) \times (-1.5)}{-2 \times -3 \times -4}$$

$$L_0 = -0.015625$$

$$L_1 = \frac{(1.5+1)(1.5-2)(1.5-3)}{(1+1)(1-2)(1-3)} = \frac{2.5 \times -0.5 \times (-1.5)}{2 \times (-1) \times (-2)}$$

$$L_1 = 0.46875$$

$$L_2 = \frac{(1.5+1)(1.5-1)(1.5-3)}{(0+2+1)(2-1)(2-3)} = \frac{2.5 \times 0.5 \times (-1.5)}{3 \times 1 \times -1}$$

$$L_2 = 0.625$$

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$$C_3 = \frac{(1.5+1)(1.5-1)(1.5-2)}{(3+1)(3-1)(3-2)}$$

$$= \frac{2.5 \times 0.5 \times (-0.5)}{4 \times 2 \times 1}$$

$$C_3 = -0.078125$$

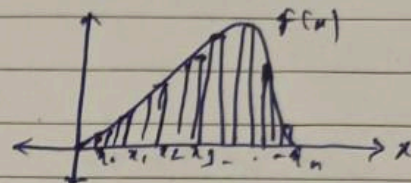
$$Y = (-0.015625)(-21) + (0.96875)(11) + (0.625)(12) + (-0.078125)(2)$$

$$Y = 14.625$$

* Numerical Differentiation and Integration:

The area of the curve $y=f(x)$ between $x(x)$ and x -axis from a to b ,

$$y = \int_a^b f(x) dx$$



Methods:

① Trapezoidal Rule $\rightarrow \int_a^b f(x) dx = h \left[\frac{y_0 + y_n}{2} + y_1 + y_2 + \dots + y_{n-1} \right]$

$N=2$

$\hookrightarrow$ It is applicable in any no. of interval even/odd.

② Simpson's $\frac{1}{3}$ rule $\rightarrow$ (It gives accurate values for quadratic polyn.)

$N=3$

$$\int_a^b f(x) dx = \frac{h}{3} \left[(y_0 + y_n) + 4(y_1 + y_3 + y_5 + y_7 + \dots) + 2(y_2 + y_4 + y_6 + \dots) \right]$$

$\rightarrow$ It is applicable when total no. of intervals is even. [e.g. $\rightarrow 0, 1, 2, 3, 4, 5, 6$]
 $\hookrightarrow$ intervals = $n-1$
 $= 6-1 = 5$

③ Simpson's $\frac{3}{8}$ rule $\rightarrow$ (It gives accurate values for cubic polyn.)

$N=3$

$$\int_a^b f(x) dx = \frac{3h}{8} \left[(y_0 + y_n) + 3(y_1 + y_2 + y_3 + y_4 + y_5 + y_6 + \dots) + 2(y_7 + y_8 + y_9 + y_{10} + \dots) \right]$$

→ It is applicable if total no. of intervals is multiple of 3.

(4) Weddle's Rule → (used for six degree polyn.)
15161525

$$\boxed{N=6} \quad \int_{x_0}^{x_6} = \frac{3h}{10} [y_0 + 5y_1 + y_2 + 6y_3 + y_4 + 5y_5 + 2y_6 + 5y_7 + \dots]$$

→ It is applicable if total no. of intervals are multiple of 6.

* $h = \frac{b-a}{n}$ where (a, b) is interval

$$n = \frac{b-a}{h}$$

Qo) Evaluate $\int_0^6 \frac{dx}{1+x^2}$ using all four methods.

Solⁿ → $a=0, b=6, f(x) = \frac{1}{1+x^2}$

Let us consider $n=6$ so that it will satisfy all conditions.

$$h = \frac{b-a}{n} = \frac{6}{6} = 1$$

x	0	1	2	3	4	5	6
f(x)	1	0.5	0.2	0.1	0.092	0.032	0.0270
					39	46	27

① Trapezoidal Rule $\rightarrow$

$$\int_0^6 \frac{dx}{1+x^2} = \frac{1}{2} \left[(1+0.027027) + 0.5 + 0.2 + 0.1 + 0.5882 \right] + 0.03846 + \cancel{0.027027}$$

$$= 1.4107 \cdot 1.4108875$$

② Simpson's $1/3$ rule $\rightarrow$

$$\int_0^6 \frac{dx}{1+x^2} = \frac{1}{3} \left[(1+0.027027) + 4(0.5 + 0.1 + 0.03846) + 2(0.2 + 0.5882) \right]$$

$$= \frac{1}{3} [1.027027 + 4(0.63846) + 2(0.7882)]$$

$$= 1.366209$$

③ Simpson's $3/8$ rule $\rightarrow$

$$\int_0^6 \frac{dx}{1+x^2} = \frac{3}{8} \left[(1+0.027027) + 3(0.5 + 0.2 + 0.05882) + 2(0.1) + 0.03846 \right]$$

$$= 1.35717$$

④

Weddle's rule $= \frac{3}{10} [y_0 + 5y_1 + y_2 + 6y_3 + y_4 + 5y_5 + 2y_6]$

$$= 1.3735$$